Sales Data Analysis Pipeline using Azure Data Factory

Objective

The objective of this project is to build a simple and automated data pipeline using Azure Data Factory (ADF). The project demonstrates how sales data can be collected, moved, and prepared for analysis in a cloud environment.

In many organizations, sales data is received in CSV files and manually transferred between different systems. This manual process can take time and increase the chances of errors. The goal of this project is to automate the movement of sales data from a source location to a processed data layer using Azure Data Factory.

This project helps in understanding the basic concepts of data engineering and cloud-based data integration, including Linked Services, Datasets, Pipelines, Activities, and Monitoring.

Project Goals

- Automate the movement of sales data.
- Learn how Azure Data Factory works.
- Understand data integration concepts.
- Create a reusable data pipeline.
- Prepare data for future reporting and analytics.
- Gain hands-on experience with Azure cloud services.

Architecture

The project uses Azure Data Factory and Azure Blob Storage to create a simple data pipeline.

Components Used

Azure Blob Storage

Azure Blob Storage is used to store the sales CSV file. It acts as the source and destination storage location.

Azure Data Factory

Azure Data Factory is used to create and manage the data pipeline. It controls the movement of data between different locations.

Linked Service

A Linked Service is used to create a connection between Azure Data Factory and Azure Blob Storage.

Dataset

Datasets represent the actual files being used in the pipeline.

Copy Activity

The Copy Activity is responsible for moving data from the source container to the processed container.

Architecture Diagram

SalesData.csv ↓ Azure Blob Storage (Source Container) ↓ Azure Data Factory ↓ Copy Activity ↓ Azure Blob Storage (Processed Container) ↓ Analytics Ready Data

This architecture demonstrates a simple Extract and Load (EL) process that is commonly used in data engineering projects.

Pipeline Flow

The pipeline follows a simple step-by-step process.

Step 1: Upload Source Data

A sample sales data file named SalesData.csv is uploaded to the source container in Azure Blob Storage.

Example Data:

OrderID, Customer, Product, Amount, Date

1001, John, Laptop, 50000, 01-06-2026

1002, Rahul, Mobile, 20000, 02-06-2026

1003, Priya, Tablet, 15000, 03-06-2026

Step 2: Create Linked Service

A Linked Service is created in Azure Data Factory to connect with Azure Blob Storage.

Purpose:

- Establish secure connection
- Allow ADF to access storage

Step 3: Create Dataset

Datasets are created for:

- Source CSV File
- Destination Processed File

Purpose:

- Define the location of data
- Define the structure of the file

Step 4: Create Pipeline

A pipeline named SalesDataPipeline is created in Azure Data Factory.

Purpose:

- Organize activities
- Control workflow execution

Step 5: Add Copy Activity

A Copy Activity is added to the pipeline.

Purpose:

- Read sales data from source container
- Copy data to processed container

Step 6: Execute Pipeline

The pipeline is executed using the Debug option.

ADF performs the following actions:

1. Connects to Blob Storage.
2. Reads the SalesData.csv file.
3. Copies the file.
4. Stores the file in the processed container.
5. Generates execution logs.

Step 7: Monitor Execution

Pipeline execution is monitored using the Monitor section in Azure Data Factory.

Information Available:

- Execution Status
- Start Time
- End Time
- Duration
- Success or Failure Messages

Dataset Details

Source Dataset

Dataset Name: DS_SalesData

Location: Azure Blob Storage

File Type: CSV (Delimited Text)

File Name: SalesData.csv

Purpose: Contains raw sales transaction data uploaded by users.

Columns:

- OrderID
- Customer
- Product
- Amount
- Date

Destination Dataset

Dataset Name: DS_ProcessedSales

Location: Azure Blob Storage

File Type: CSV (Delimited Text)

Purpose: Stores processed sales data after pipeline execution.

Benefits:

- Centralized storage
- Ready for analytics
- Easy integration with reporting tools

Results

The project successfully demonstrates how Azure Data Factory can automate the movement of sales data within a cloud environment.

Achievements

✔ Created Azure Blob Storage containers.

✔ Created Linked Services to establish connectivity.

✔ Created source and destination datasets.

✔ Built a data pipeline using Azure Data Factory.

✔ Implemented Copy Activity for automated data movement.

✔ Executed and monitored pipeline runs successfully.

✔ Prepared sales data for future reporting and analytics.

Skills Learned

Through this project, the following skills were developed:

- Azure Data Factory Fundamentals
- Linked Services
- Datasets
- Pipelines
- Copy Activity
- Azure Blob Storage
- Data Integration
- Pipeline Monitoring
- Cloud Data Engineering Concepts

Business Benefits

This solution helps organizations by:

- Reducing manual work
- Improving data availability
- Supporting data analytics initiatives
- Increasing process efficiency
- Providing a scalable cloud-based solution

Conclusion

This project provides a practical introduction to Azure Data Factory and cloud-based data integration. It demonstrates how sales data can be automatically moved and prepared for analysis using Azure services. The project serves as a strong foundation for learning more advanced topics such as Data Flows, Azure Databricks, Microsoft Fabric, Power BI, and Data Engineering.